Syed Muhammad Abuzar Rizvi

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Summary

Ph.D. candidate specializing in quantum computing, quantum machine learning, hybrid quantum–classical algorithms, and quantum communication, with a strong publication record in peer-reviewed journals and international conference proceedings. Google Scholar (Oct. 2026): 87 citations, h-index 5, i10-index 3.

Skills

- Programming Skills & Tools: C/C++, Python, TensorFlow, Keras, Qiskit, PennyLane, QuTiP, CPU/GPU-based quantum simulations, MATLAB, LaTeX.
- Experienced with quantum computing on various modalities, quantum machine learning, quantum circuits, variational quantum algorithms, secure quantum communication, and fundamental quantum information science theory.
- 5 years of leadership experience crafting research proposals, project management, and mentoring researchers.
- Ability to collaborate with other team members and work independently to solve complex problems.

Experience

- **Kyung Hee University (CQILAB)** *Mar. 2021–Present*
  Graduate Research Associate Yongin-si, South Korea
  - Developed machine-learning-based frameworks for quantum state classification, state discrimination, and quantum error mitigation on noisy intermediate-scale quantum devices.
  - Engineered variational quantum circuits on IBM Quantum backends for hybrid quantum–classical optimization, including VQE- and QUBO-based formulations for network scheduling and NP-hard decision problems.
  - Designed and evaluated quantum algorithms for next-generation intelligent networks, with applications in edge intelligence, resource optimization, and secure quantum communication.
  - Investigated reinforcement-learning-based quantum control, pulse-level gate design, and anonymous quantum semantic communication protocols for reliable and privacy-preserving quantum networks.
  - Led and contributed to high-impact journal manuscript preparation, including technical writing, result analysis, figure design, reviewer-response preparation, and submissions to top-tier journals.
  - Contributed to funded research programs through proposal writing, project planning, technical documentation, progress reports, and research outcome preparation.
  - Supported lab operations by coordinating research activities, mentoring junior students, assisting new student admissions and onboarding, and guiding students in experiments, implementation, and academic writing.
- **2025 IonQ Summer Research Mentoring Program (QCenter, SeeQI Research Group)** *Summer 2025*
  Online Research Intern Online
  - Investigated explainability in quantum machine learning under Dr. Oliver Knitter, focusing on how data encoding, entangling gates, and circuit depth affect model interpretability. Program organized by the Quantum Information Research Support Center (QCenter).
- **Microwave Engineering Research Lab (NUST)** *Oct. 2018–Feb. 2021*
  DSP Engineer Rawalpindi, Pakistan
  - Architected hardware–software co-design solutions on Xilinx Zynq-7000 SoC platforms, programming the embedded ARM Cortex-A9 processor in C for low-latency hardware control and real-time data orchestration.
  - Engineered high-throughput RTL modules in Verilog and VHDL using Vivado, implementing real-time DSP pipelines for FIR filtering, FFT processing, fixed-point conversion, and DDR memory interfacing.
  - Designed and integrated high-speed mixed-signal processing interfaces for ADC/DAC data acquisition, SMA-based signal routing, and clock synchronization in microwave system testbeds.
  - Translated signal-processing algorithms into deployable FPGA-based hardware implementations, ensuring robust system integration, timing closure, and reliable real-time performance.

Education

- **Kyung Hee University (KHU)** *Mar. 2021–Present*
  Ph.D. in Electronics and Information Convergence Engineering Yongin-si, South Korea
  - GPA: 4.09/4.30
- **National University of Sciences and Technology (NUST)** *Sep. 2014–Jun. 2018*
  Bachelor of Science in Electrical Engineering Rawalpindi, Pakistan
  - GPA: 3.50/4.00

Proposal Writing & Project Experience

[Self Principal Investigator]

- Hybrid Quantum-Classical Computation for Optimization and Machine Learning
  Funding Agency: Brain Korea 21, South Korea (2021–2022)
- Variational Quantum Computing for Optimization: Bridging Quantum Computing, Machine Learning, and Quantum Algorithms
  Funding Agency: Brain Korea 21, South Korea (2023–2024)

[Team Member]

- Convergence Engineering Institute for Emerging Communication Technology
  Funding Agency: National Research Foundation of Korea (2021–2023)
- Development of 6H Next-Generation Mobile Communications Technology
  Funding Agency: Institute of Information & Communications Technology Planning & Evaluation (2021–2026)
- Foundations for Quantum Networking with Quantum Enigmata
  Funding Agency: National Research Foundation of Korea (2021–2024)
- Wi-Qi BRL: Toward Integrating 6G Wireless and 2G Quantum
  Funding Agency: National Research Foundation of Korea (2022–2025)
- BK21 Education and Research Group for Innovative Human Resources in Convergence Future Communications
  Funding Agency: National Research Foundation of Korea (2022–2027)
- Graduate School of Convergence Security, Kyung Hee University
  Funding Agency: Institute of Information & Communications Technology Planning & Evaluation (2024–2025)
- KGCC International Joint Research Group for Computing-Converged Next-Generation Communications
  Funding Agency: Kyung Hee University (2023–2024)
- Foundations for Computing-Intensive Quantum Communications: Toward Meta-Privacy
  Funding Agency: National Research Foundation of Korea (2025–2027)
- Research Promotion and Competitiveness Enhancement of the Quantum Information Technology Application Research Institute, Global Software Convergence Research Institute
  Funding Agency: Kyung Hee University (2025–2026)

Publications and Patents

Journal Papers

1. J.U. Rehman, N. Iqbal, **S.M.A. Rizvi**, H. Shin, "Quantum Random Access Codes for Image Transmission," *Scientific Reports*, 2026.
2. U.I. Paracha, **S.M.A. Rizvi**, U. Khalid, T.Q. Duong, M.Z. Win, H. Shin, "Quantum Fusion Learning for 6G Networks with Kolmogorov-Arnold Networks," *IEEE Transactions on Network Science and Engineering*, 2026. (IF = 7.9)
3. A. Khan, **S.M.A. Rizvi**, T.Q. Duong, E.K. Hong, H. Shin, "Quantum Full-Duplex Anonymous Communication in Adversarial Quantum Networks," *IEEE Transactions on Network Science and Engineering*, 2026. (IF = 7.9)
4. H.P.Y. Duong, **S.M.A. Rizvi**, B. McNiven, T.T. Nguyen, H. Shin, O.A. Dobre, T.Q. Duong, "Quantum Machine Learning for Drug Discovery: Taxonomy, Research Challenges, and the Road Ahead," *ACM Computing Surveys*, vol. 58, no. 8, pp. 1–36, 2026. (IF = 28.0)
5. U. Khalid, U.I. Paracha, **S.M.A. Rizvi**, H. Shin, "Quantum Computing for Intelligent Transportation Systems: VQE-Based Traffic Routing and EV Charging Scheduling," *Mathematics*, vol. 13, no. 17, pp. 2761, 2025. (IF = 2.2)
6. **S.M.A. Rizvi**, U.I. Paracha, U. Khalid, K. Lee, H. Shin, "Quantum Machine Learning: Towards Hybrid Quantum-Classical Vision Models," *Mathematics*, vol. 13, no. 16, pp. 2645, 2025. (IF = 2.2)
7. **S.M.A. Rizvi**, U. Khalid, S. Chatzinotas, T.Q. Duong, H. Shin, "Controlled Quantum Semantic Communication for Industrial CPS Networks," *IEEE Transactions on Network Science and Engineering*, vol. 13, pp. 996–1009, 2025. (IF = 7.9)
8. J.U. Rehman, **S.M.A. Rizvi**, S. Koudia, S. Chatzinotas, H. Shin, "MIMO Quantum Communication in Correlated Pure-Loss Channels," *IEEE Communications Letters*, vol. 29, no. 7, pp. 1544–1548, 2025. (IF = 4.4)
9. **S.M.A. Rizvi**, N. Asif, M.S. Ulum, T.Q. Duong, H. Shin, "Multiclass Classification of Metrologically Resourceful Tripartite Quantum States with Deep Neural Networks," *Sensors*, vol. 22, no. 18, pp. 6767, 2022. (IF = 3.5)

Conference Proceedings

1. **S.M.A. Rizvi**, J.U. Roh, H. Shin, "Deep Reinforcement Learning for Optimal Pulse Shaping in Single-Photon Quantum Emitters," in *40th International Conference on Information Networking (ICOIN)*, pp. 180–184, 2026.

2. E. Gholamalishahiharouni, A.W. Shaffar, **S.M.A. Rizvi**, U. Khalid, H. Shin, "Analytical Design of an LC Double-Notch Purcell Filter for Superconducting Quantum Circuits," in *Korea Information and Communications Society Conference Proceedings*, pp. 330–331, Jeju, South Korea, Jun. 2025. **Best Paper Award**
3. M.M.U. Gondel, **S.M.A. Rizvi**, U. Khalid, H. Shin, "NISQ State Preparation with Reinforcement Learning," in *Korea Information and Communications Society Conference Proceedings*, pp. 950–951, Jeju, South Korea, Jun. 2025.
4. U.I. Paracha, **S.M.A. Rizvi**, M.M.U. Gondel, W. Park, H. Shin, "Adaptive Quantum Readout Error Mitigation with Transfer Learning," in *15th International Conference on Information and Communication Technology Convergence*, Jeju, South Korea, Oct. 2024.
5. **S.M.A. Rizvi**, M.M.U. Gondel, U.I. Paracha, H. Shin, "Variational Quantum Eigensolver for Optimizing Network Scheduling Using QUBO Formulation," in *International Conference on Industrial Networks and Intelligent Systems*, pp. 46–57, Da Nang, Vietnam, Feb. 2024.
6. **S.M.A. Rizvi**, M.S. Ulum, N. Asif, H. Shin, "Neural Networks with Variational Quantum Circuits," in *International Conference on Industrial Networks and Intelligent Systems*, pp. 203–214, Ho Chi Minh City, Vietnam, Aug. 2023. **Best Paper Award**
7. **S.M.A. Rizvi**, F. Zaman, K. Lee, H. Shin, "Quantum Optimal Control with Pulses," in *Korea Information and Communications Society Conference Proceedings*, pp. 153–154, Gyeongju, South Korea, Nov. 2022.
8. **S.M.A. Rizvi**, M.A. Ullah, H. Shin, "Measurement Error Mitigation for NISQ Devices," in *Korea Information and Communications Society Conference Proceedings*, pp. 1393–1394, Jeju, South Korea, Jun. 2022.
9. F. Zaman, **S.M.A. Rizvi**, H. Shin, "Quantum-Assisted Reinforcement Learning," in *Korea Information and Communications Society Conference Proceedings*, pp. 516–518, Jeju, South Korea, Jun. 2022.
10. **S.M.A. Rizvi**, M.A. Ullah, H. Shin, "Machine Learning for Qubit State Estimation on IBM Quantum Computer," in *Korea Information and Communications Society Conference Proceedings*, pp. 472–474, Pyeongchang, South Korea, Feb. 2022. **Best Paper Award**

Patents

1. H. Shin, **S.M.A. Rizvi**, F. Dzulfikar, M.S. Ulum, J.U. Roh, "Apparatus and Method of Mitigating Errors in Quantum Fourier Transform," *Korea Patent Application No. 10-2023-0158507* [submitted].
2. H. Shin, H.J. Jung, M.S. Choi, F. Zaman, **S.M.A. Rizvi**, S.A. Wachid, F. Dzulfikar, "Method of Error Mitigation for Noisy Intermediate-Scale Quantum (NISQ) Computation," *Korea Patent Application No. 10-2023-0002208* [submitted].
3. H. Shin, J.U. Roh, U. Khalid, **S.M.A. Rizvi**, T. Muhammad, "Quantum Reservoir Learning for Reinforcement-Learning-Optimized Pulse and Hamiltonian Control System and Method for a Rydberg Lattice," *Korea Patent Application No. 10-2026-0044766* [submitted].
4. H. Shin, **S.M.A. Rizvi**, A. Naema, U. Khalid, M.A. Ullah, A. Khan, "Apparatus and Method for Classifying Quantum States in Quantum Information Integrated Processing System," *Korea Patent No. 10-2820473*.

Honors, Awards and Scholarships

- Best Paper Awards: Korea Information and Communications Society Conference (2022, 2025); International Conference on Industrial Networks and Intelligent Systems (2023).
- KHU Khreative U+ Fellowship winner, 2021 and 2023.
- Brain Korea 21 (BK21) Plus Doctoral Research Scholarship, Kyung Hee University, 2021–Present.
- Kyung Hee University Graduate Student Scholarship, 2021–Present.
- NUST GPA-based merit scholarship.

Certifications

- **Qiskit Global Summer School on Quantum Machine Learning** *July 2021*
- **2025 IonQ Summer Research Mentoring Program, Certificate of Participation (QCenter)** *2025*

References

Available upon request.